

1) What is HTML? What does HTML stand for?

- HTML stands for Hypertext Markup Language.
- HTML is used to create and structure web pages.

It tells the browser how the content of a webpage should be displayed.

2) Why is HTML called a markup language? Is HTML a Programming language?

- HTML is called a markup language because it uses tags to mark and structure content on a webpage.

For example:

`<h> welcome </h>`

`<p> This is a Paragraph </p>`

- HTML is not a programming language because it does not perform logical operations like loops, conditions, or calculations. It is mainly used to structure web content.

3) What is the difference between an HTML tag and an HTML Element?

HTML tag:

A tag the markup written inside angle brackets

Example:

`<h> HTML`

`<p>`

`</p>`

HTML Element:-

An HTML element includes the opening tags, content, and closing tag.

Example:-

<HTML

<P> Hello world </P>

Here:-

- <P> → opening tag
- Hello world → Content
- </P> → closing tag
- Entire <P> Hello world </P> → HTML Element.

4) What is <!DOCTYPE html>? Is it an HTML tag?

- <!DOCTYPE html> is a declaration that tells the browser that the document is an HTML5 document.
- It should be placed at the top of an HTML document.

Example:-

<HTML

<!DOCTYPE html>

No, it is not an HTML tag. It is a declaration.

5) Explain the basic structure of an HTML document. What are the purpose of <html>, <head>, and <body>?

Basic HTML structure:-

<HTML

<!DOCTYPE html>

<html>

<head>

<title> my website </title>

</head>

<body>

<h1> Hello </h1>

<p> welcome to my website. </p>

</body>

</html>

<html>

- It is the root element of the HTML document.
- All HTML content is placed inside it.

<head>

- It contains information about the web page, such as

- title
- meta information
- CSS links
- JavaScript links

<body>

- It contains all the visible content of the webpage, such as.

- Headings
- Paragraphs
- Images
- Buttons
- Links

6) What are HTML attributes? Explain with an example.

HTML attributes provide additional information about HTML element.

They are usually written inside the opening tag.

Example:-

<!DOCTYPE html>

 click here

Here:-

- href → Attribute

- "https://example.com" → Attribute value

Another example:-

<!DOCTYPE html>

src and alt are attributes.

7) What is the difference between `src` and `href`?

- `src`

- `src` stands for source.

- It is used to embed or load a resource into a webpage.

Example:-

- `href`

- `href` stands for Hypertext Reference.

- It is used to create a link to another webpage or resource.

Example:-

 Google

 Google

Difference:-

- `src` → loads a resource.

- `href` → creates a link to another resource

8) What is the purpose of the `alt` attribute in an `img` tag?

- The `alt` attribute provides alternative text for an image.

- It is displayed or used when the image cannot be loaded and helps screen readers understand the image.

Example:-

9) Explain the difference between Block-level elements and Inline elements with examples.

Block-level elements:-

• Block-level elements start on a new line and usually take the full available width.

Example:-

```
<1> HTML
```

```
<div>
```

```
<p>
```

```
<h1>
```

Ex:-

```
<1> HTML
```

```
<div> Hello </div>
```

```
<div> world </div>
```

• These will appear on separate lines.

Inline elements:-

Inline elements do not start on a new line. They only take the space required by their content.

Examples:-

```
<1> HTML
```

```
<span>
```

```
<a>
```

```
<strong>
```

Example:-

```
<1> HTML
```

```
<span> Hello </span>
```

```
<span> word </span>
```

These can appear on the same line.

10) What is Semantic HTML? what is the difference between Semantic and non-Semantic elements?

Semantic HTML :-

Semantic elements clearly describe the meaning of their content.

Examples :-

<header>

<nav>

<article>

<section>

<footer>

For example :-

< HTML

<header>

<h1> my website </h1>

</header>

Non-Semantic HTML :-

Non-Semantic elements do not describe the meaning of their content.

Examples :-

< HTML

<div>

Difference :-

- Semantic elements → describe the meaning of the content
- Non-Semantic elements → do not describe the meaning of the content.

11) What is the difference between `<div>` and `<span>`?

`<div>`:

- Block-level element
- Starts on a new line
- Used to group larger section of content

Example:

```
<!DOCTYPE HTML>
<div>
  <h1> Hello </h1>
  <p> Welcome </p>
</div>
```

`<span>`:

- Inline element
- Does not start on a new line
- Used to style or group small parts of text

Example:

```
<!DOCTYPE HTML>
<p> Hello <span> world </span> </p>
```

12) What are the different types of HTML lists? Explain `<ul>`, `<ol>`, and `<dl>`.

- There are three main types of HTML lists.

i) `<ul>` - unordered list

- Display items using bullets

```
<!DOCTYPE HTML>
<ul>
  <li> Apple </li>
  <li> Mango </li>
  <li> Orange </li>
</ul>
```


(i) `<ol>` - ordered list

- Display items using numbers.

`<ol>` HTML

`<ol>`

`<li>` Apple `</li>`

`<li>` mango `</li>`

`<li>` Orange `</li>`

`</ol>`

(ii) `<dl>` - Description list.

- used for items and their descriptions.

`<dl>` HTML

`<dl>`

`<dt>` HTML `</dt>`

`<dd>` Hyperlink markup language `</dd>`

`</dl>`

13) Explain the important HTML table tag

i) `<table>` ::

- The `<table>` tag is used to create a table in HTML

Syntax :-

`<table>` HTML

`<table>`

Table content

`</table>`

ii) `<tr>`

- The `<tr>` tag stands for Table Row - it is used to create a row in a table.

Syntax :-

`<tr>`

Table cells

`</tr>`

iv) `<th>`

The `<th>` tag stands for table header. It is used to create heading cells in a table.

Example:-

```
<th> Name </th>
```

```
<th> Age </th>
```

v) `<td>`

The `<td>` tag stands for table data. It is used to display normal data inside a table.

Example:-

```
<td> vijay </td>
```

```
<td> 21 </td>
```

vi) `<thead>`

The `<thead>` tag is used to group the header content of a table.

Example:-

```
<thead>
```

```
<tr>
```

```
<th> Name </th>
```

```
<th> Age </th>
```

```
</tr>
```

```
</thead>
```

vii) `<tbody>`

The `<tbody>` tag is used to group the main data or body content of a table.

Example:-

```
<tbody>
```

```
<tr>
```

```
<td> vijay </td>
```

```
<td> 21 </td>
```

```
</tr>
```

```
</tbody>
```

vii) <tfoot>

The <tfoot> tag is used to group the footer content of a table.

Example:

```
<tfoot>
  <tr>
    <td> Total </td>
  </tr>
</tfoot>
```

Complete Table Example:

```
<table border="1">
  <thead>
    <tr>
      <th> Name </th>
      <th> Age </th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td> vijay </td>
      <td> 21 </td>
    </tr>
  </tbody>
  <tfoot>
    <tr>
      <td> Total students </td>
      <td> 1 </td>
    </tr>
  </tfoot>
</table>
```


14) What is the purpose of <meta charset="UTF-8"> and the view port meta tag?

• meta charset = "UTF-8"

• The <meta> charset = "UTF-8" tag is used to specify the character encoding of an HTML document.

• UTF-8 allows the webpage to correctly display:

- English characters
- Tamil characters
- other languages
- special symbols

Syntax:

<meta charset="UTF-8">

view port meta tag:

The view port meta tag helps a webpage display properly on mobile phones, tablets, and different screen sizes.

Syntax:

<meta name="viewport" content="width=device-width, initial-scale=1.0">

Meaning:

• width = device-width → The webpage width adjust to the device screen width.

• Initial-scale = 1.0 → Set the initial zoom level.

Example:-

<head>

<meta charset="UTF-8">

<meta name="viewport" content="width

=device-width, initial-scale=1.0">

</head>

In short:-

- The meta tags help display character correctly and make the website mobile-friendly.

15) What is open Graph (OG)? Explain the purpose of og:title, og:type, og:image, og:url, and og:description.

What is open Graph (OG)?

- open Graph (OG) is a set of meta tags used to control how a web page appears when it is shared on social media platforms.

It can display information such as:-

- Title

- Description

- Image

- Website link

Example:-

<meta property="og:title" content="my website">

<meta property="og:type" content="website">

<meta property="og:image" content="image.jpg">

<meta property="og:url" content="https://example.com">

<meta property="og:description" content="welcome to my website">

i) og:title:-

• It specifies the title of the web page when shared on social media.

• Example:

<meta property="og:title" content="my Portfolio">

ii) og:type:-

It specifies the types of content

Examples include:

- website
- article
- video

Example:

<meta property="og:type" content="website">

ii) og: image :-

• It specifies the image that should be displayed when the webpage is shared

Example:-

<meta property = "og: image" content = "image.jpg"

iv) og: url :-

• It specifies URL link in the webpage

Example:-

<meta property = "og: url" content = "https://"

Example.com"

v) og: description

• It specifies a short description of the webpage

Example:-

<meta property = "og: description" content = "welcome to my personal portfolio website"